Data Independent Acquisition (DIA)

Proteomic short Course 2025

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Before DIA

- Profiling was done by DDA
  - Spectral counting
  - LFQ
  - TMT (10 or 11 plex)

Example from a Paper we published in Plant Cell in 2019 using Spectral Counts and MS1 LFQ!

A

Yeast/Mammals/Arabidopsis	FET	NC r1	NC r2	NC r3	ROG2 r1	ROG2 r2	ROG2 r3
		Spectral counts					
- /TRAPPC11/ROG2 (*bait*)	<0.0001	0	0	0	259	273	285
TRS20/TRAPPC2/At2g20930	<0.0001	0	0	0	12	6	14
TRS23/TRAPPC4/At5g02280	<0.0001	0	0	0	5	3	4
TRS31/TRAPPC5/At5g58030	<0.0001	0	0	0	7	9	12
TRS33/TRAPPC6/At3g05000	<0.0001	0	0	0	20	20	20
BET3/TRAPPC3/At5g54750	<0.0001	0	0	0	48	42	38
TRS85/TRAPPC8/At5g16280	<0.0001	0	0	0	92	85	81
TRS120/TRAPPC9/At5g11040		nd	nd	nd	nd	nd	nd
TRS130/TRAPPC10/At5g54440		nd	nd	nd	nd	nd	nd

B

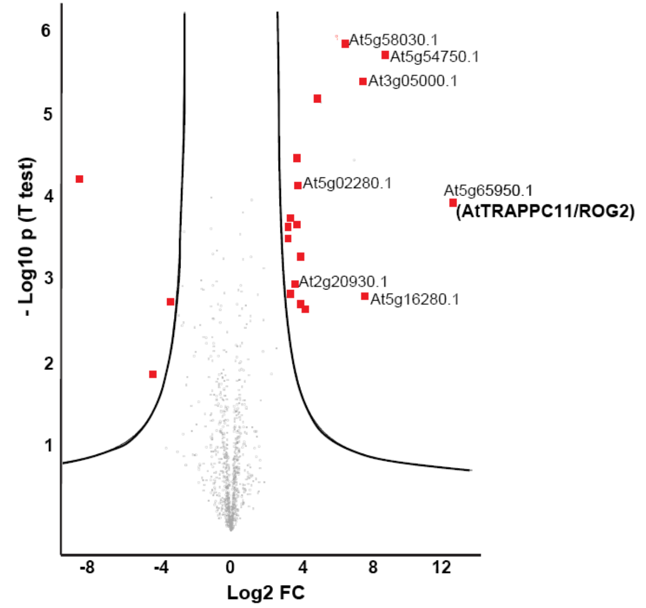
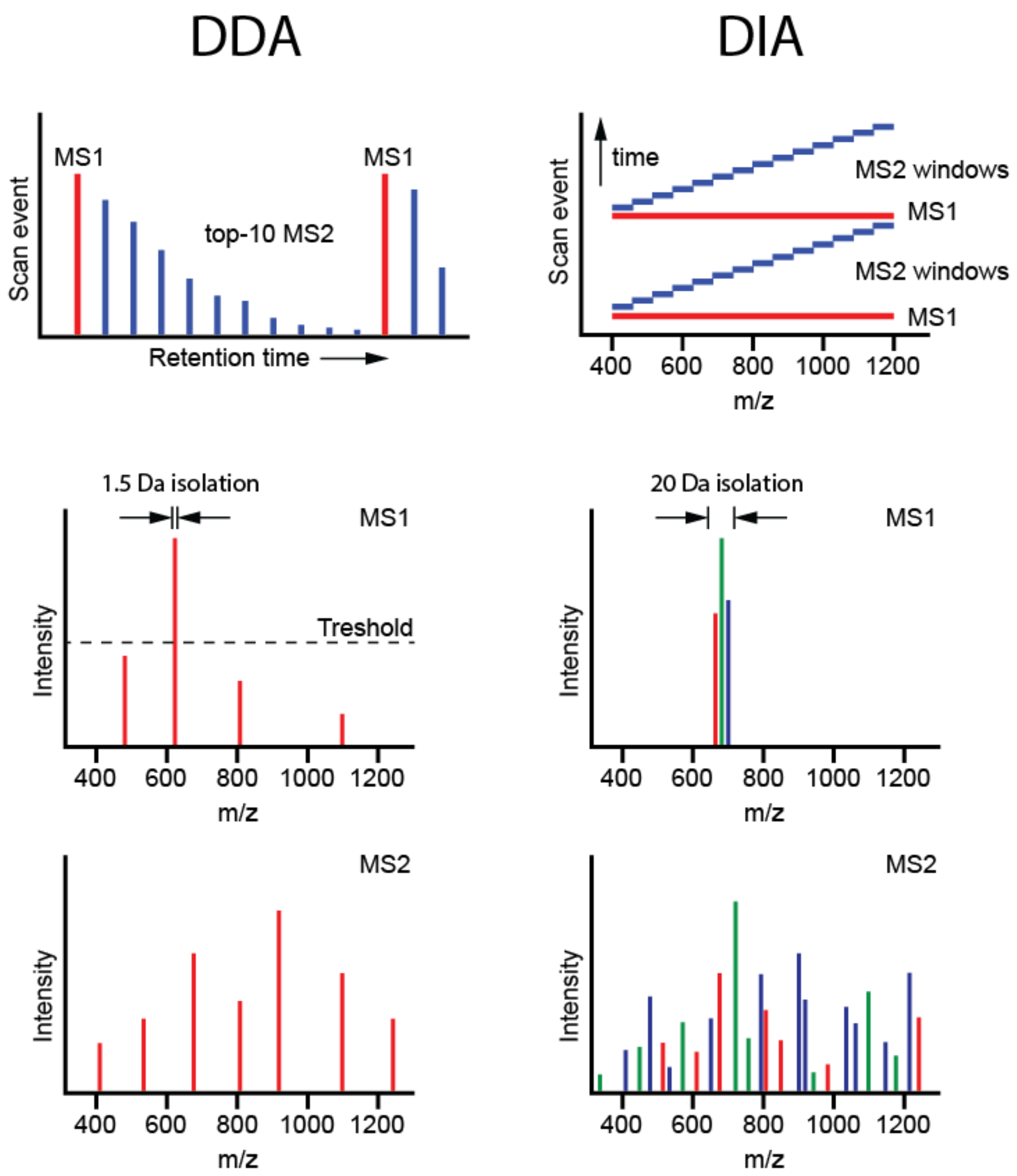


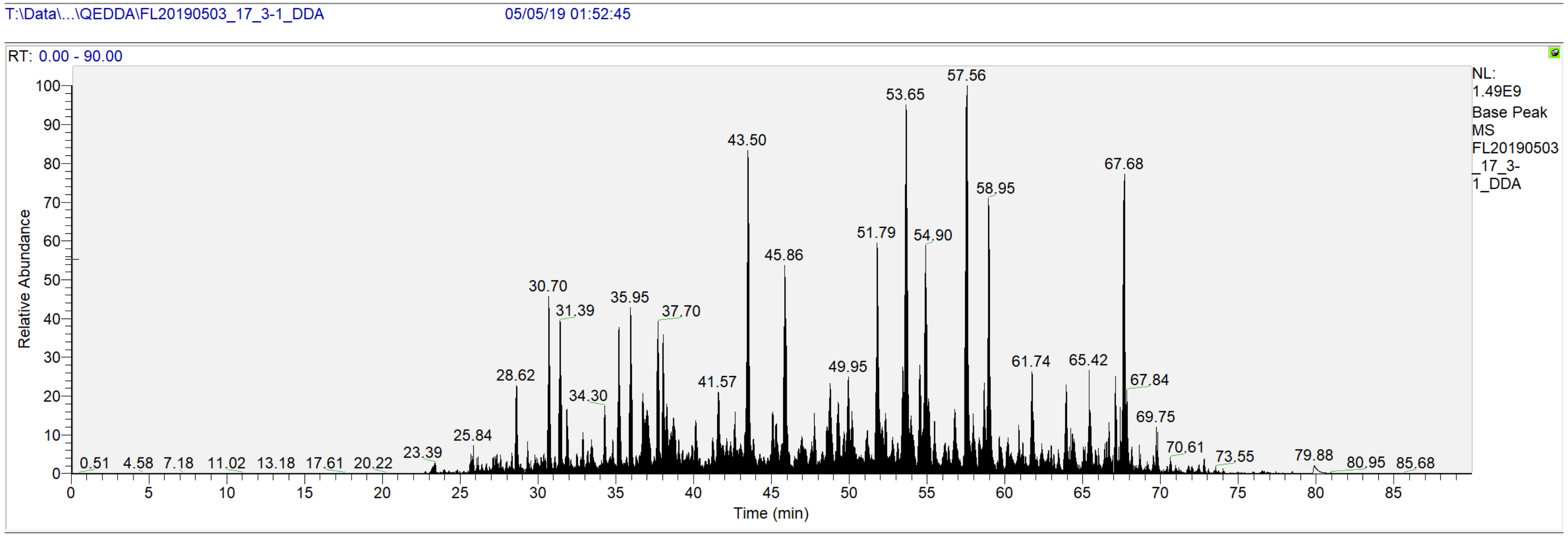
Figure 7. AtTRAPPC11/ROG2 defines a novel Arabidopsis TRAPP complex.

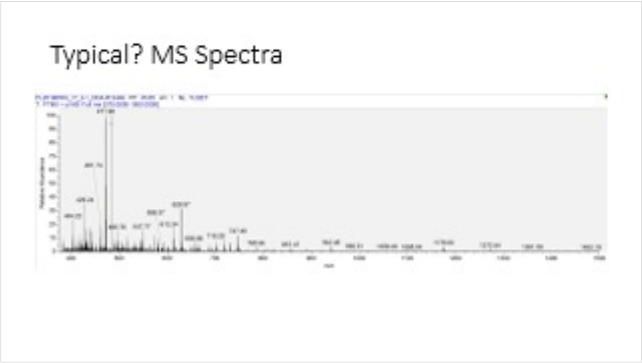
(A) Spectral counts are presented for the six Arabidopsis TRAPP orthologs significantly enriched in YFP-ROG2 immunoprecipitates ("ROG2 r1-r3"), and for AtTRAPPC11/ROG2 itself (bait). r: biological replicate (1-3). FET: Fisher's exact test. NC: negative control. Note that Arabidpopsis orthologs of TRS120 and TRS130, defining the TRAPPII complex, were not detected (nd) in our analysis.
(B) Volcano plot illustrating log2 fold change (FC) (x axis) and statistical significance distribution (y axis) of the proteomic data set (LFQ intensities). Red squares indicate proteins significantly enriched in YFP-ROG2 immunoprecipitates. Analyses were performed in triplicates.

DDA vs DIA

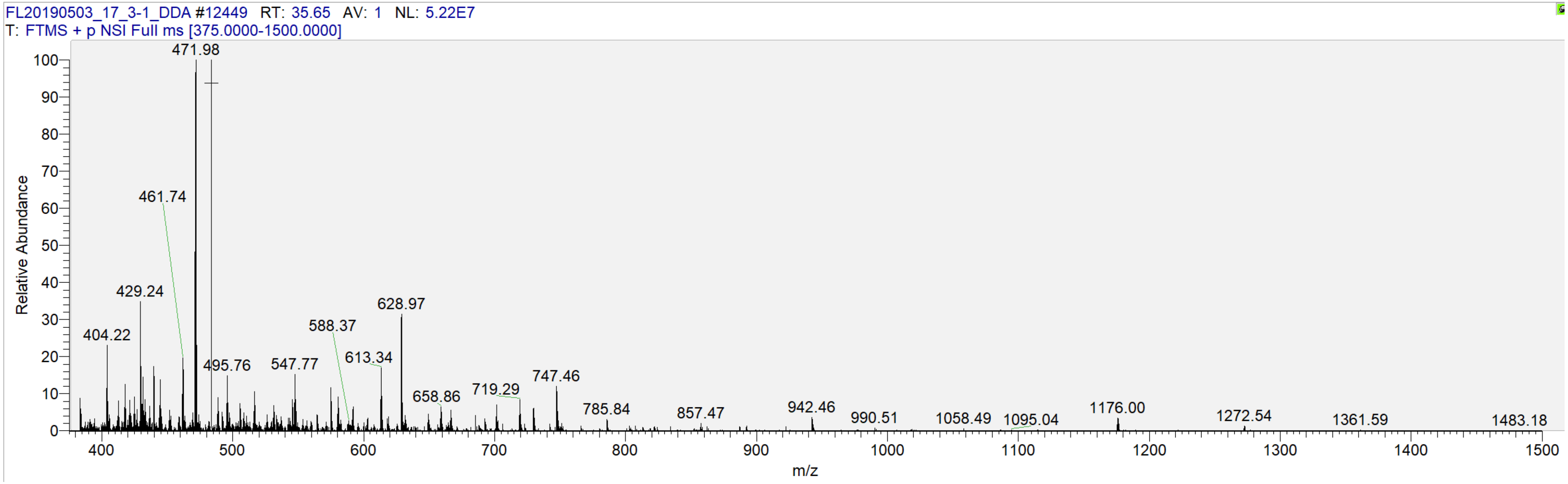

DDA
DIA
MS1
MS1
top-10 MS2
Scan event
Retention time
time
MS2 windows
MS1
MS2 windows
MS1
Scan event
400 600 800 1000 1200
m/z
1.5 Da isolation
MS1
Treshold
Intensity
400 600 800 1000 1200
m/z
20 Da isolation
MS1
Intensity
400 600 800 1000 1200
m/z
MS2
Intensity
400 600 800 1000 1200
m/z
MS2
Intensity
400 600 800 1000 1200
m/z

In addition to the semi randomness of DDA. Peak picking is an issue with MS1 LFQ

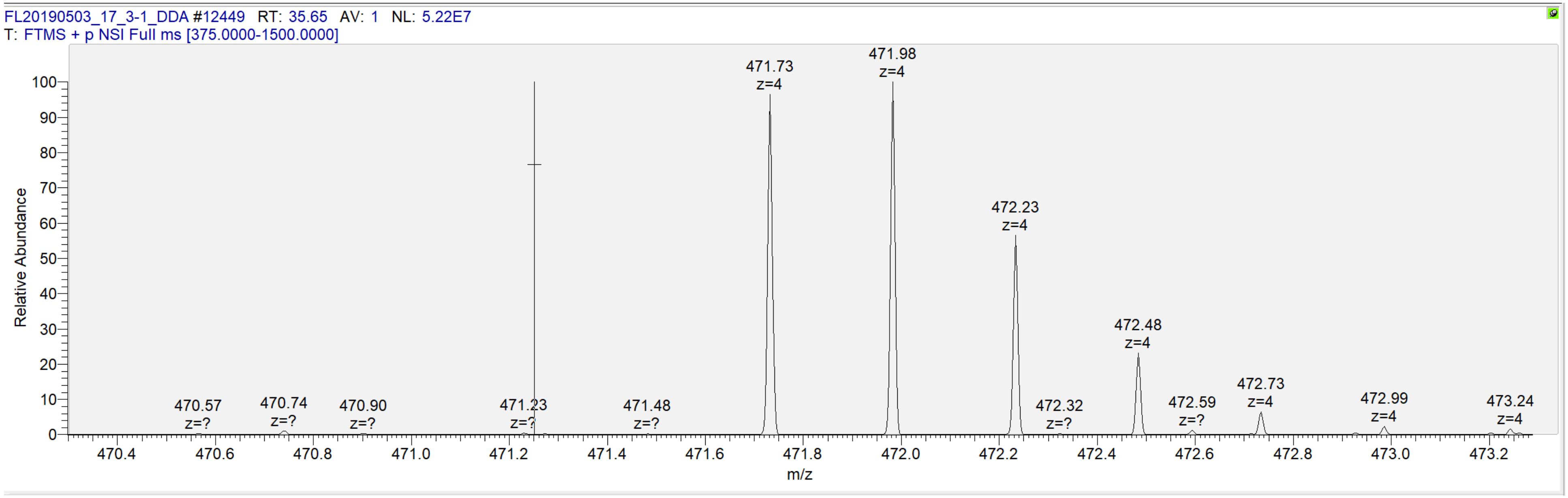




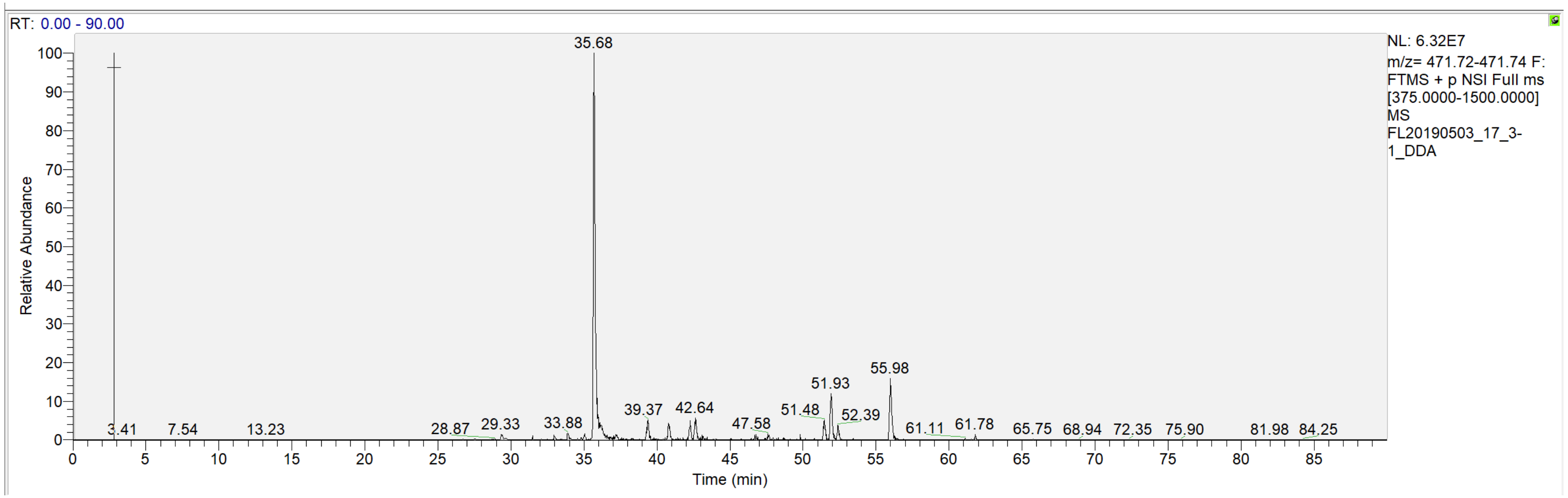
Typical? MS Spectra



Nice S/N

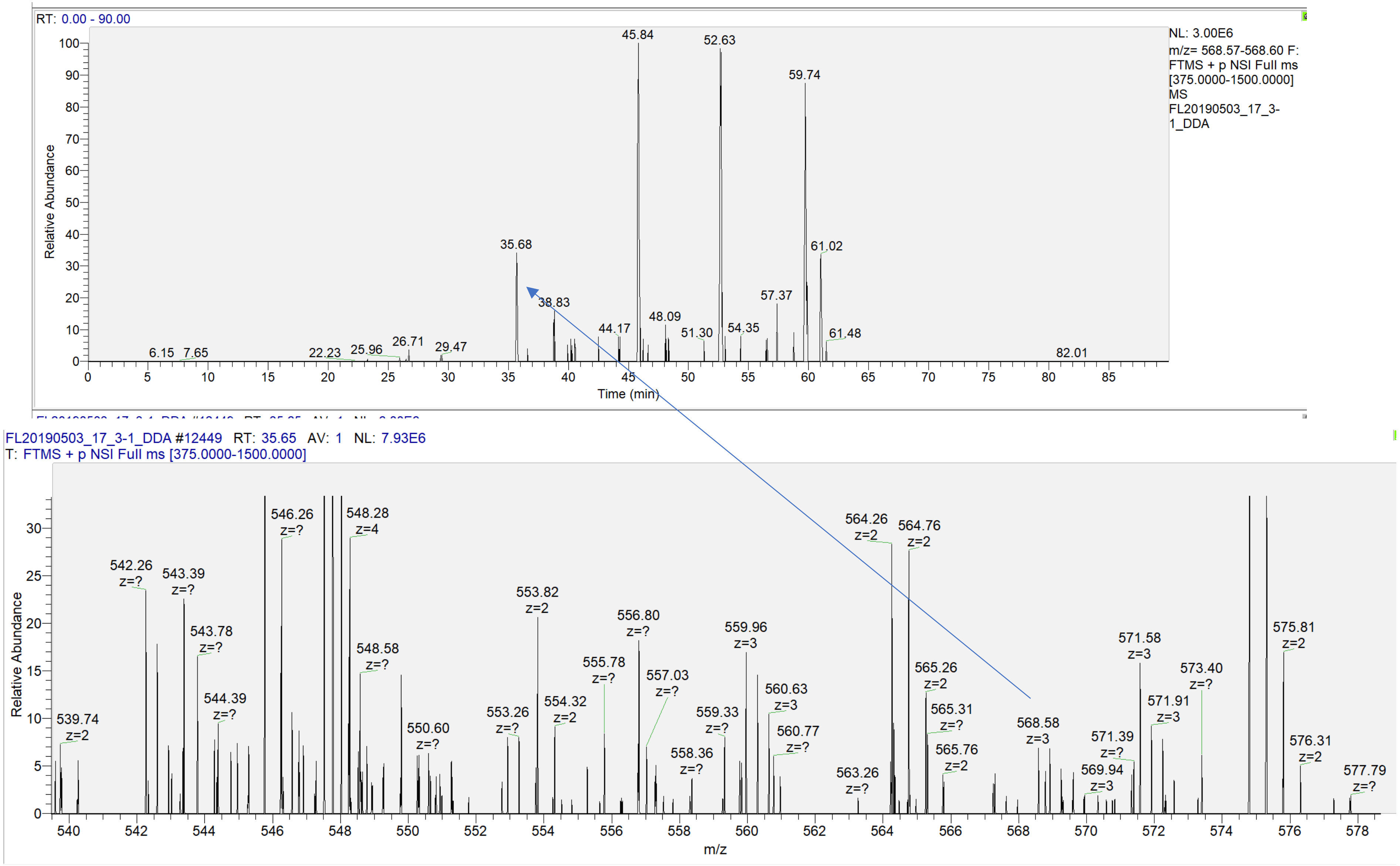


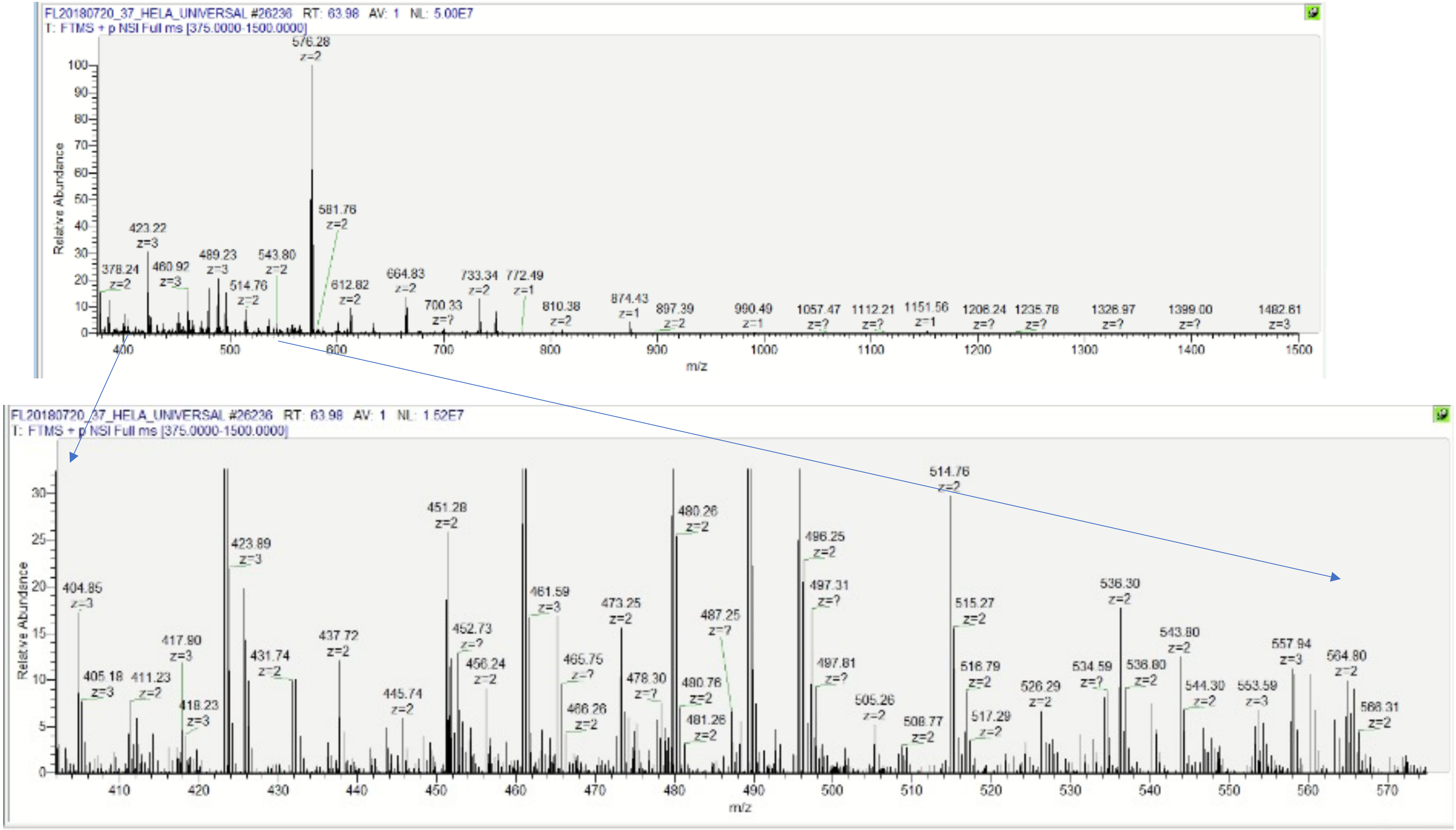
FL20190503_17_3-1_DDA #12449 RT: 35.65 AV: 1 NL: 5.22E7
T: FTMS + p NSI Full ms [375.0000-1500.0000]
Relative Abundance
m/z
471.73 z=4
471.98 z=4
472.23 z=4
472.48 z=4
472.73 z=4
472.99 z=4
473.24 z=4
470.57 z=?
470.74 z=?
470.90 z=?
471.23 z=?
471.48 z=?
472.32 z=?
472.59 z=?



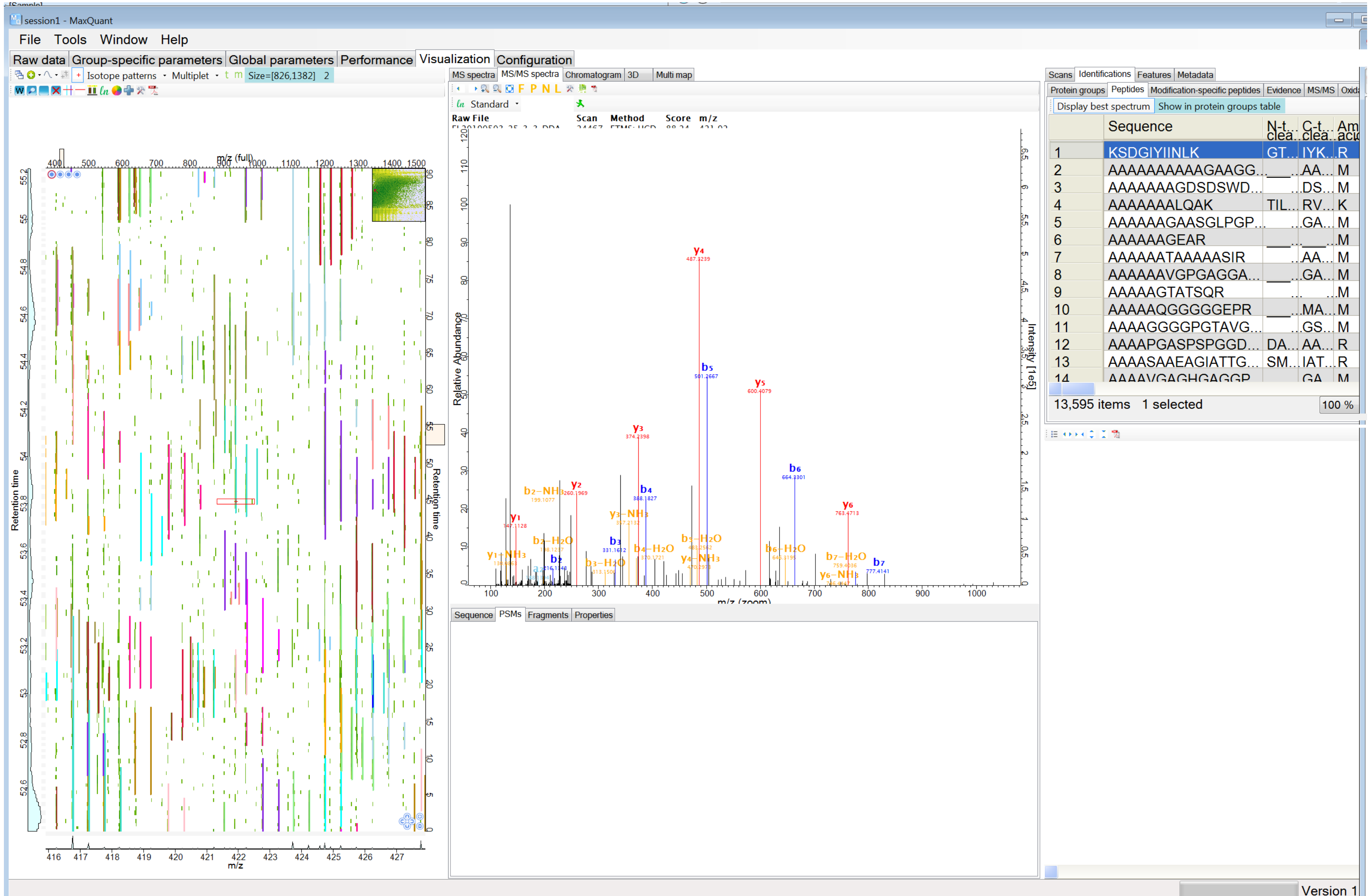
RT: 0.00 - 90.00
NL: 6.32E7
m/z= 471.72-471.74 F: FTMS + p NSI Full ms [375.0000-1500.0000] MS FL20190503_17_3-1_DDA
Relative Abundance
Time (min)
35.68
3.41
7.54
13.23
28.87
29.33
33.88
39.37
42.64
47.58
51.48
51.93
52.39
55.98
61.11
61.78
65.75
68.94
72.35
75.90
81.98
84.25

Lots of MS1 of peptides looks like this





FL20180720_37_HELA_UNIVERSAL #26236 RT: 63.98 AV: 1 NL: 5.00E7
T: FTMS + p NSI Full ms [375.0000-1500.0000]
Relative Abundance
m/z
576.28 z=2
581.76 z=2
423.22 z=3
378.24 z=2
460.92 z=3
489.23 z=3
543.80 z=2
514.76 z=2
612.82 z=2
664.83 z=2
700.33 z=?
733.34 z=2
772.49 z=1
810.38 z=2
874.43 z=1
897.39 z=2
990.49 z=1
1057.47 z=?
1112.21 z=?
1151.56 z=1
1206.24 z=?
1235.78 z=?
1326.97 z=?
1399.00 z=?
1482.61 z=3
FL20180720_37_HELA_UNIVERSAL #26236 RT: 63.98 AV: 1 NL: 1.52E7
T: FTMS + p NSI Full ms [375.0000-1500.0000]
Relative Abundance
m/z
404.85 z=3
405.18 z=3
411.23 z=2
417.90 z=3
418.23 z=3
423.89 z=3
431.74 z=2
437.72 z=2
445.74 z=2
451.28 z=2
452.73 z=?
456.24 z=2
461.59 z=3
465.75 z=?
466.26 z=2
473.25 z=2
478.30 z=?
480.26 z=2
480.76 z=2
481.26 z=2
487.25 z=?
496.25 z=2
497.31 z=?
497.81 z=?
505.26 z=2
508.77 z=2
514.76 z=2
515.27 z=2
516.79 z=2
517.29 z=2
526.29 z=2
534.59 z=?
536.30 z=2
536.80 z=2
543.80 z=2
544.30 z=2
553.59 z=3
557.94 z=3
564.80 z=2
568.31 z=2


session1 - MaxQuant
File
Tools
Window
Help
Raw data
Group-specific parameters
Global parameters
Performance
Visualization
Configuration
Isotope patterns
Multiplet
Size=[826,1382]
MS spectra
MS/MS spectra
Chromatogram
3D
Multi map
Standard
Raw File
Scan
Method
Score
m/z
m/z (full)
Retention time
m/z
Relative Abundance
Intensity [1e5]
m/z (zoom)
Sequence
PSMs
Fragments
Properties
Scans
Identifications
Features
Metadata
Protein groups
Peptides
Modification-specific peptides
Evidence
MS/MS
Display best spectrum
Show in protein groups table
Sequence
1 KSDGIYIINLK
2 AAAAAAAAAAGAAGG...
3 AAAAAAAGDSDSWD...
4 AAAAAAALQAK
5 AAAAAAGAASGLPGP...
6 AAAAAAGEAR
7 AAAAAATAAAAASIR
8 AAAAAAVGPGAGGA...
9 AAAAAGTATSQR
10 AAAAAQGGGGGEPR
11 AAAAGGGGPGTAVG...
12 AAAAPGASPSPGGD...
13 AAAASAAEAGIATTG...
13,595 items
1 selected
100 %

Missing Values Comparison Peptide level

- DDA (maxquant 1% FDR protein and peptide)
- Fusion Lumos

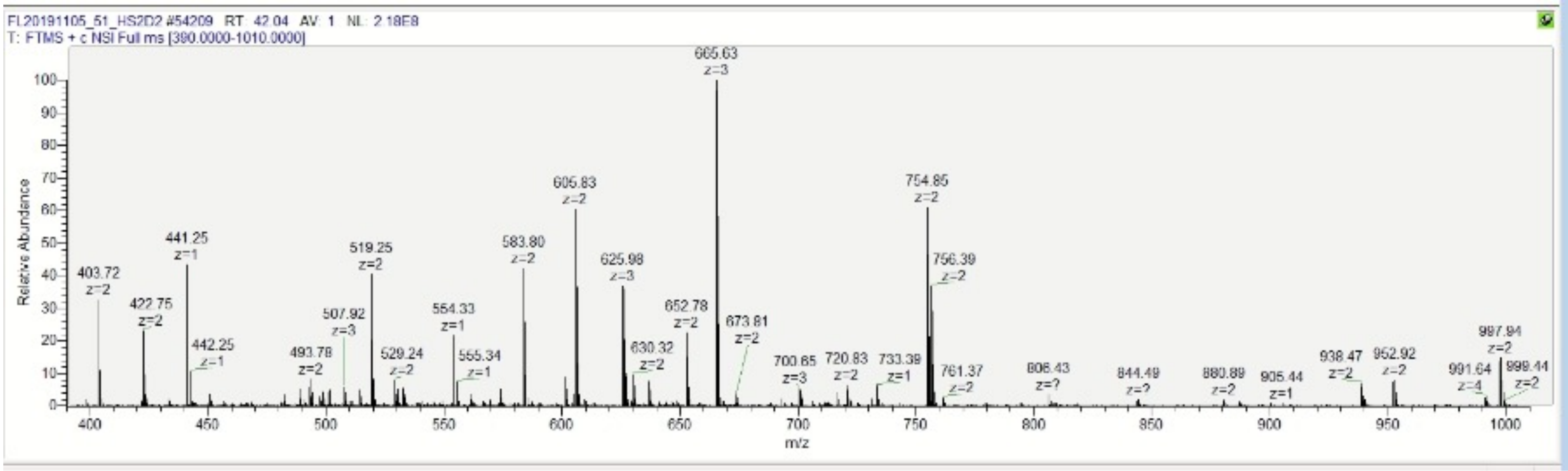
Replicate	Missing Values
1	16.3%
2	15.15%
3	23.47%

Missing in all 3 = 5.53%

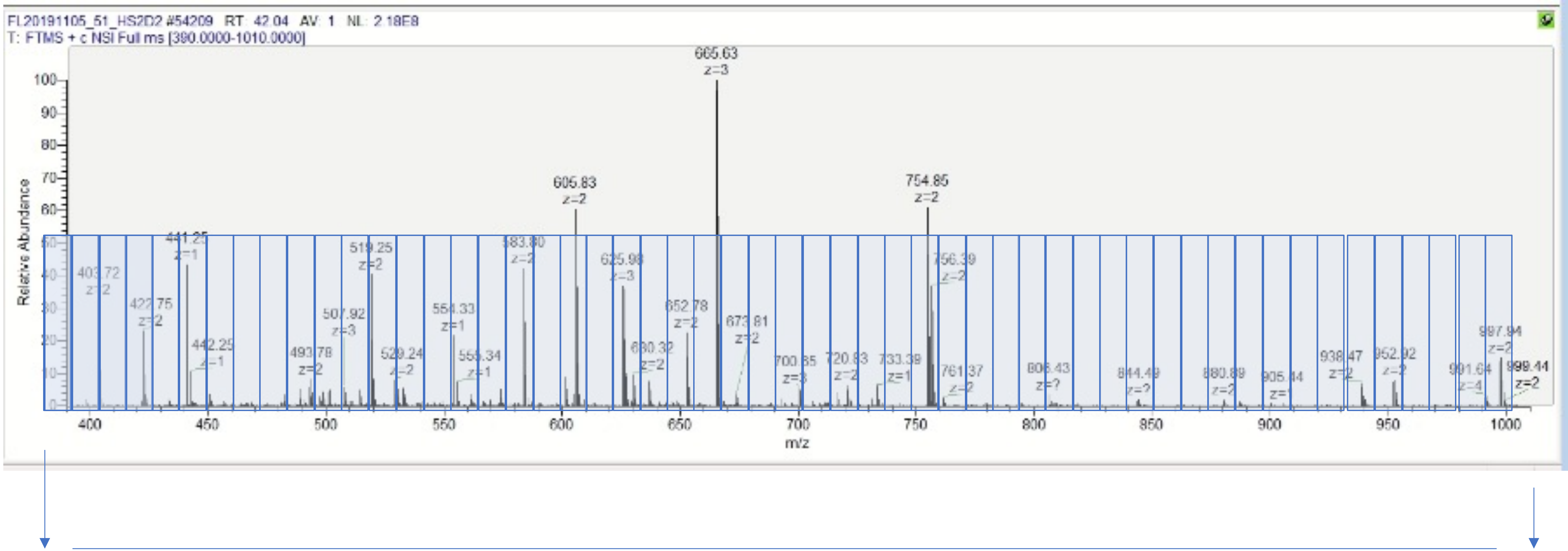
- In my hands Even with MBR on, it's difficult to get consistent peak picking across multiple samples

Real DIA example

Example MS1 Spectrum

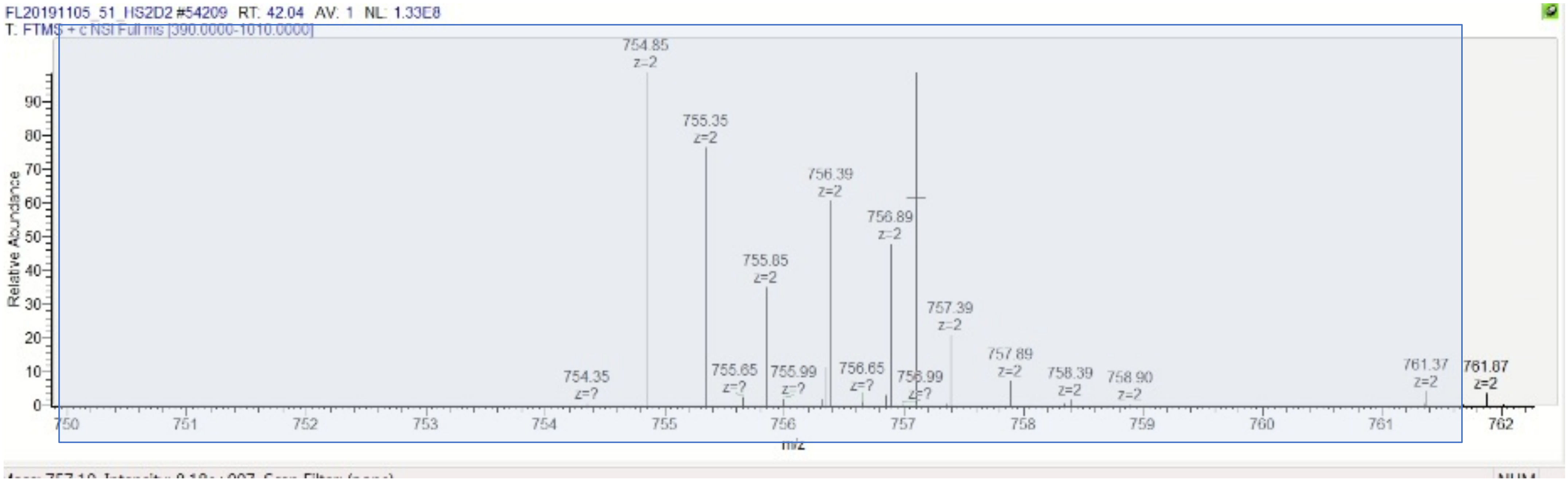


Real Example MS1 Spectrum

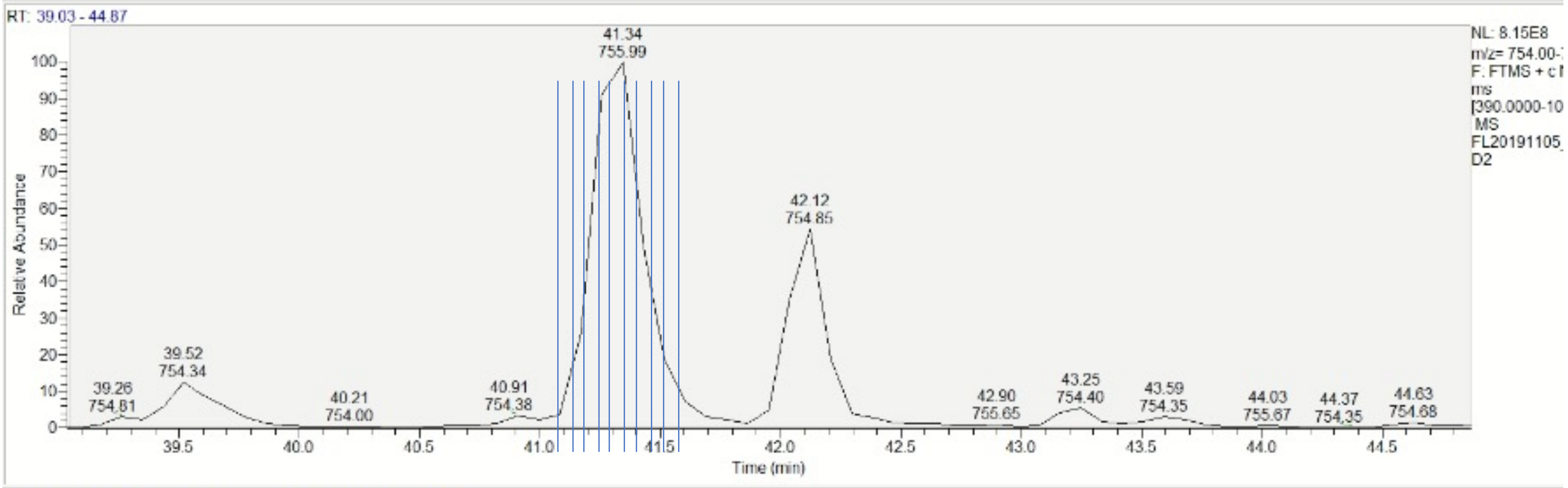


1.5-2 sec total time

MS1 spectra zoomed in to around 750 m/z

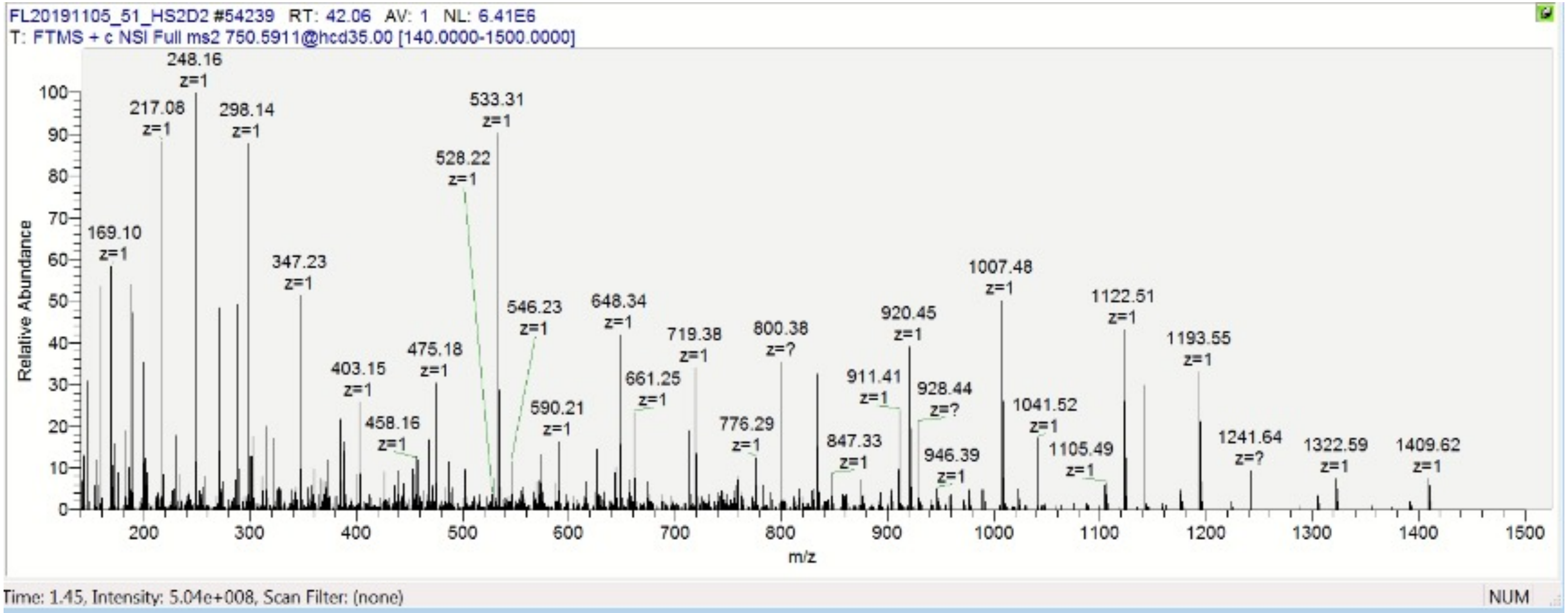


T:\service\...\FL20191105_51_HS2D2 11/06/19 09:07:04

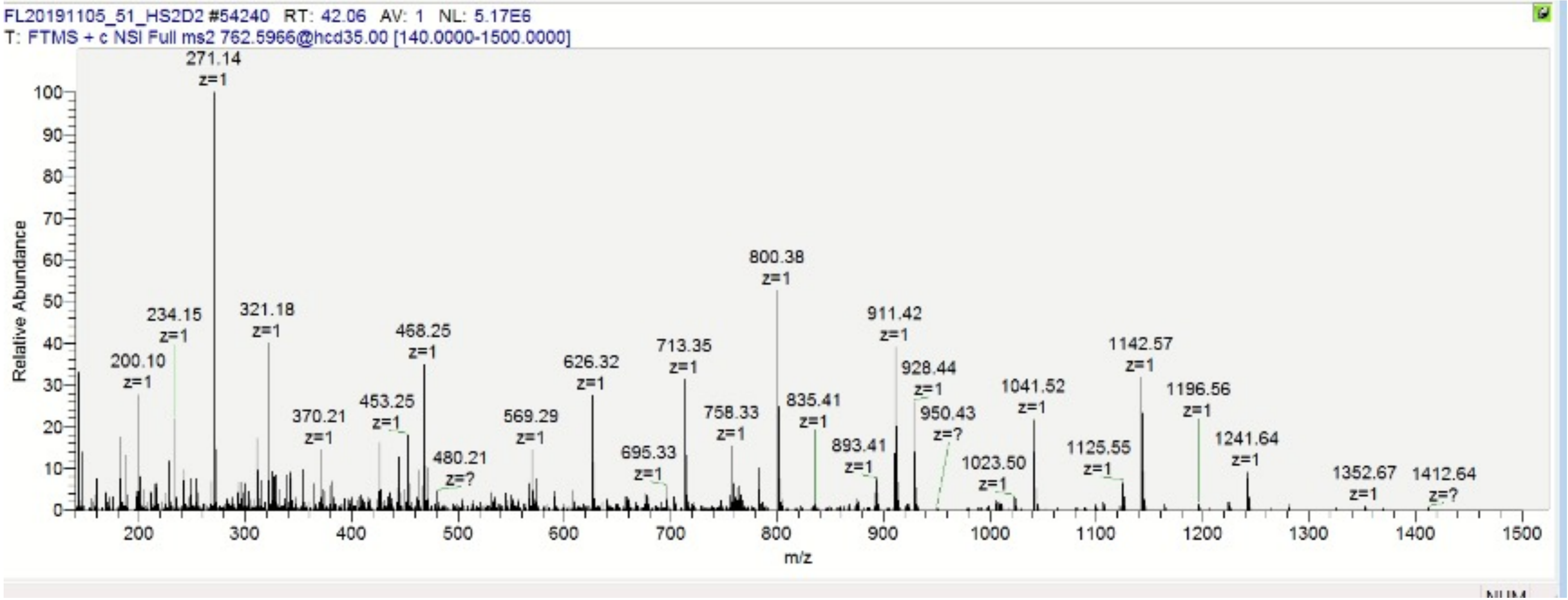


FL20191105_51_HS2D2 #54209 RT: 42.04 AV: 1 NL: 1.33E8

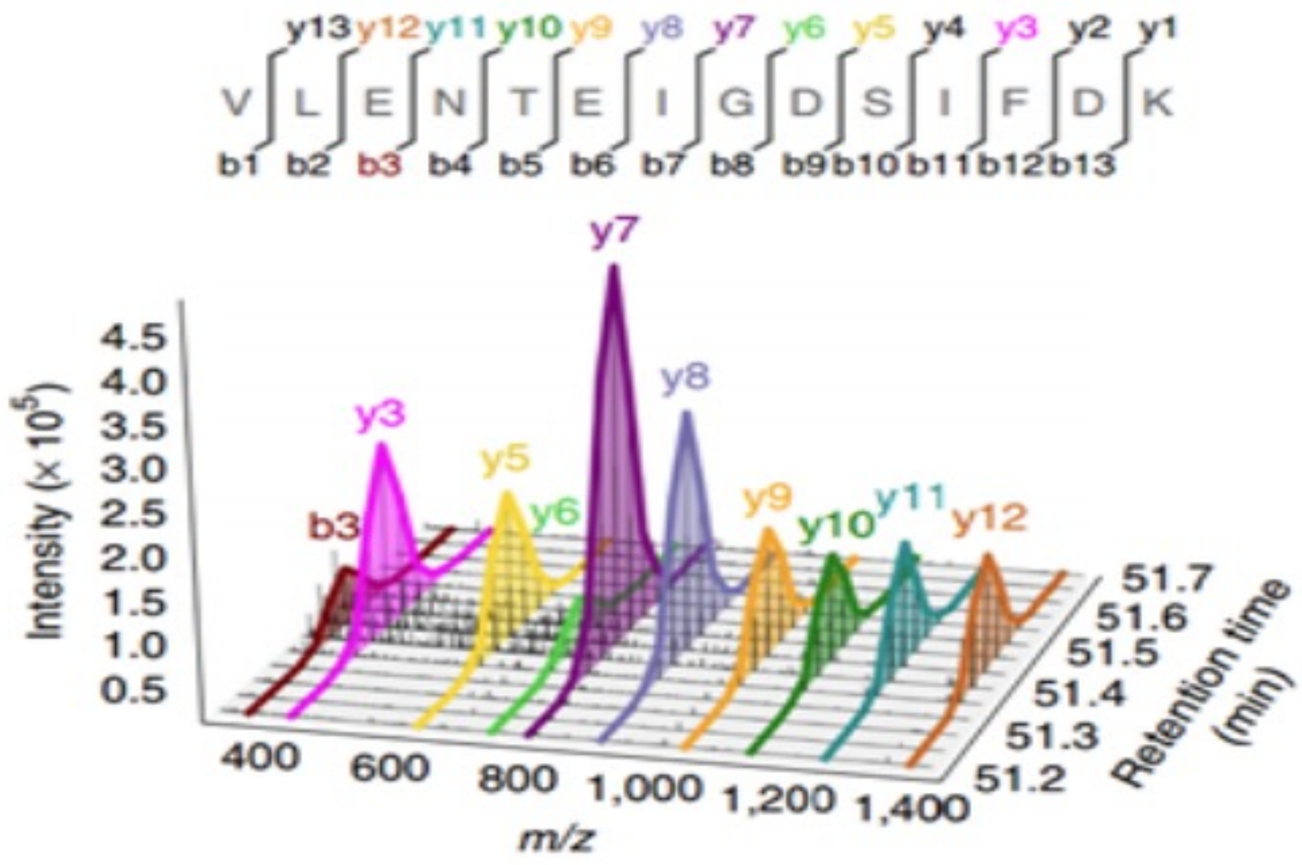
Real Example DIA spectrum 12 Da Window 750.5911 m/z



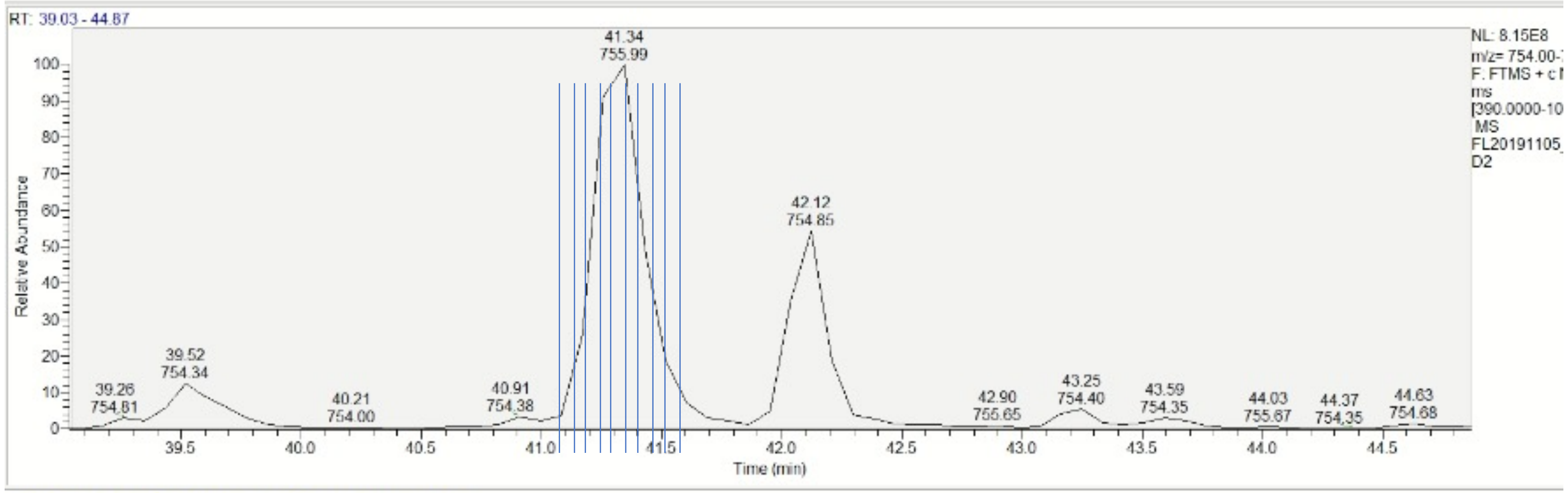
Next DIA spectrum 12 Da Window 762.5966 m/z



Remember LC-MS is three dimensional! You can use Retention time to check what fragment ions elute at the same time

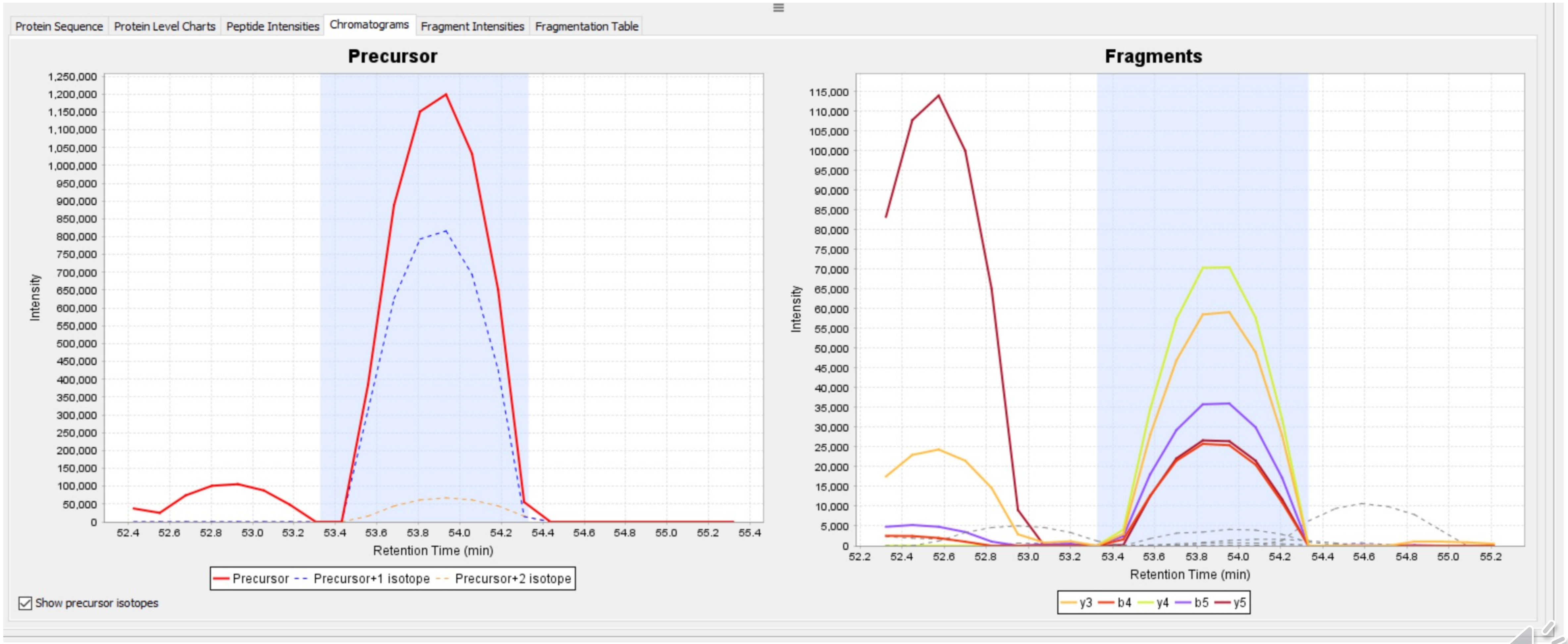


T:\service\...\FL20191105_51_HS2D2 11/06/19 09:07:04

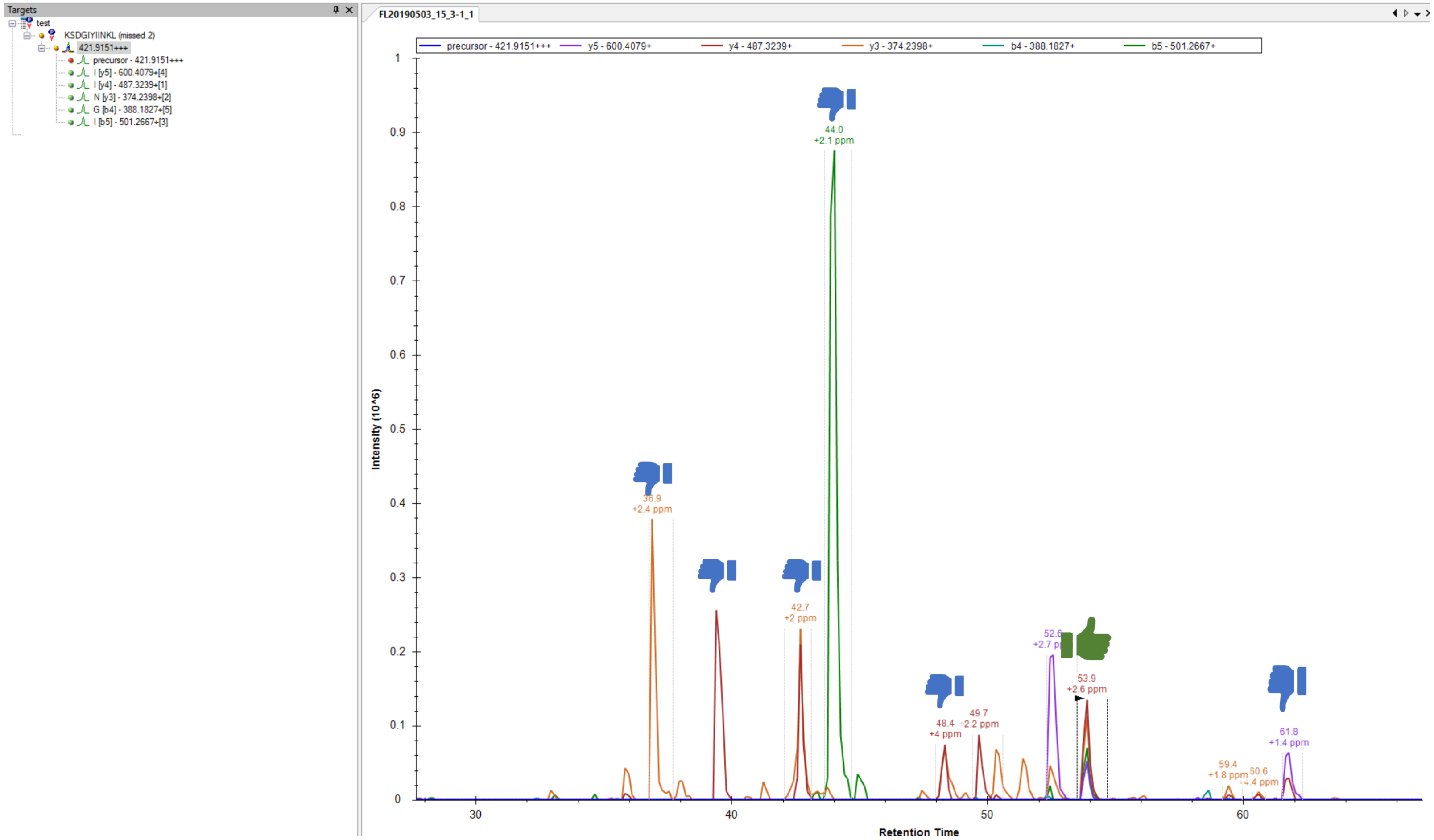


FL20191105_51_HS2D2 #54209 RT: 42.04 AV: 1 NL: 1.33E8

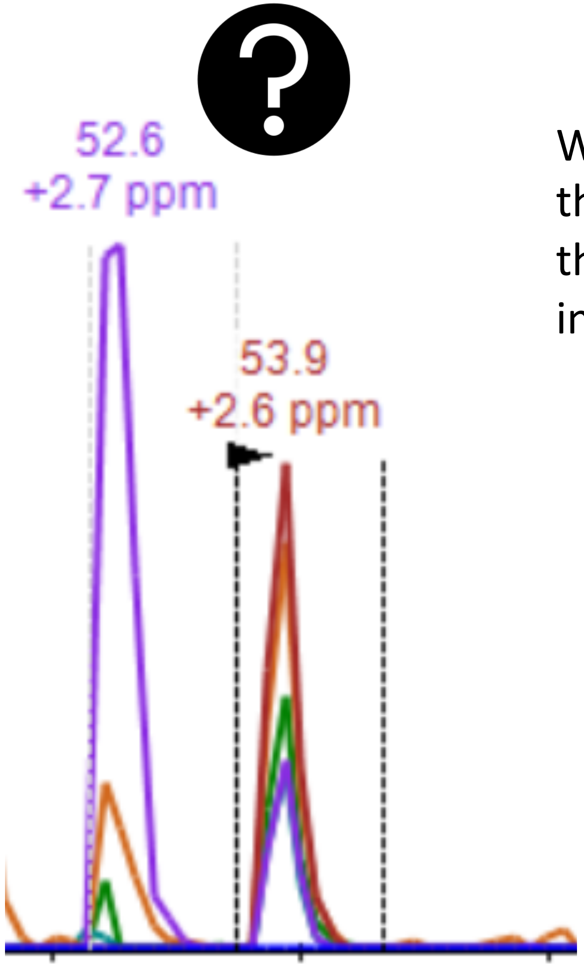
Example Peptide Centric detection



DIA Random peptide KSDGIYIINLK

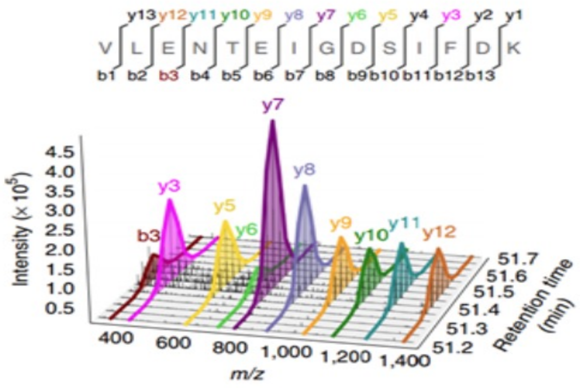


DIA Random peptide KSDGIYIINLK

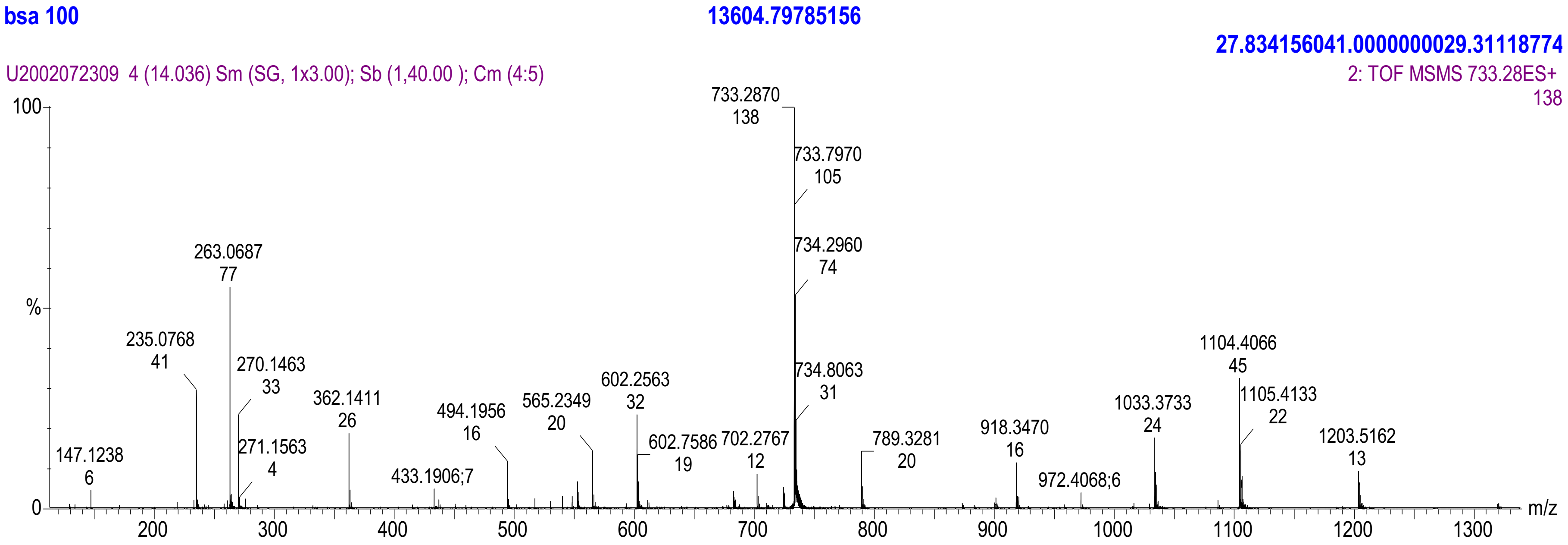


Why is the one on the right correct and the one on the left incorrect?

Check with DDA Data (DDA library)
Check with Deep learning (Prosit)



- DDA Library Spectra



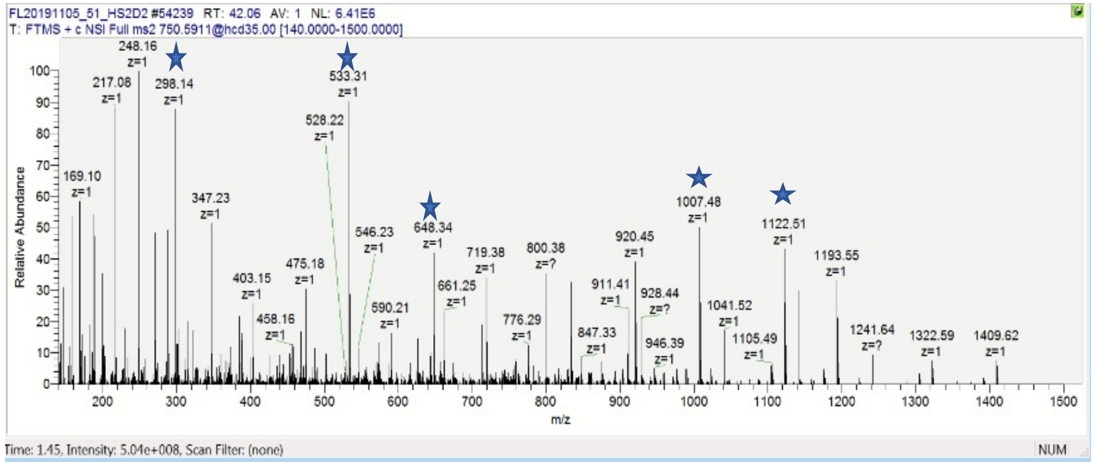
Many different ways to analyze DIA

- Spectrum Centric
  - Fragpipe
- Reconstruct DDA spectrum from DIA Data by using RT
- Match Spectra using traditional DDA methods

Some say this scales better for large search spaces

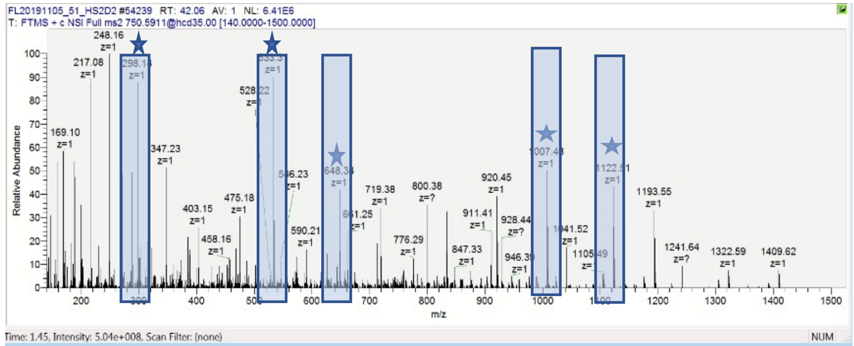
- Peptide Centric
  - Spectronaut
  - DIA-NN
- Tests every theoretical peptide in a proteome or subset of a proteome against the DIA Data to see if it's there.
  - Uses Deep learning to model theoretical peptide characterizes
    - RT
    - Fragment intensity
    - Fragment intensity ratios
- It's all filtering and fragment / precursor correlation

Spectrum Centric

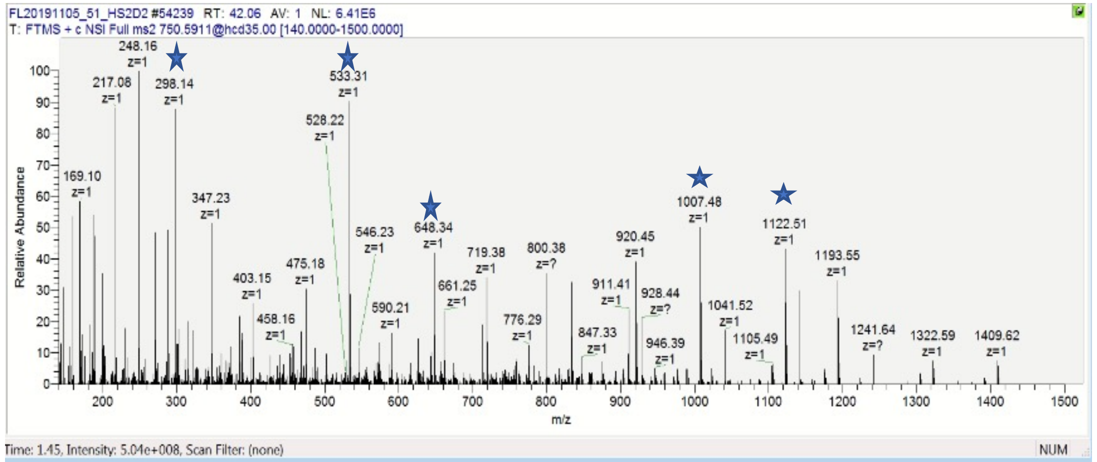


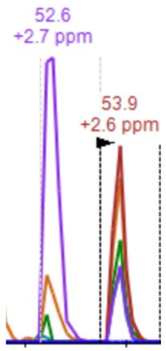
Makes a DDA spectra from ions correlated in time

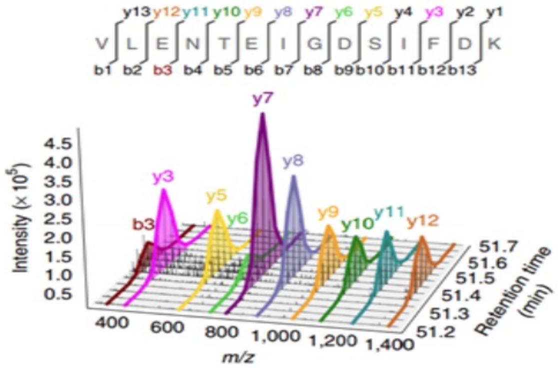
Does PSM like traditional search engine against peptide sequence



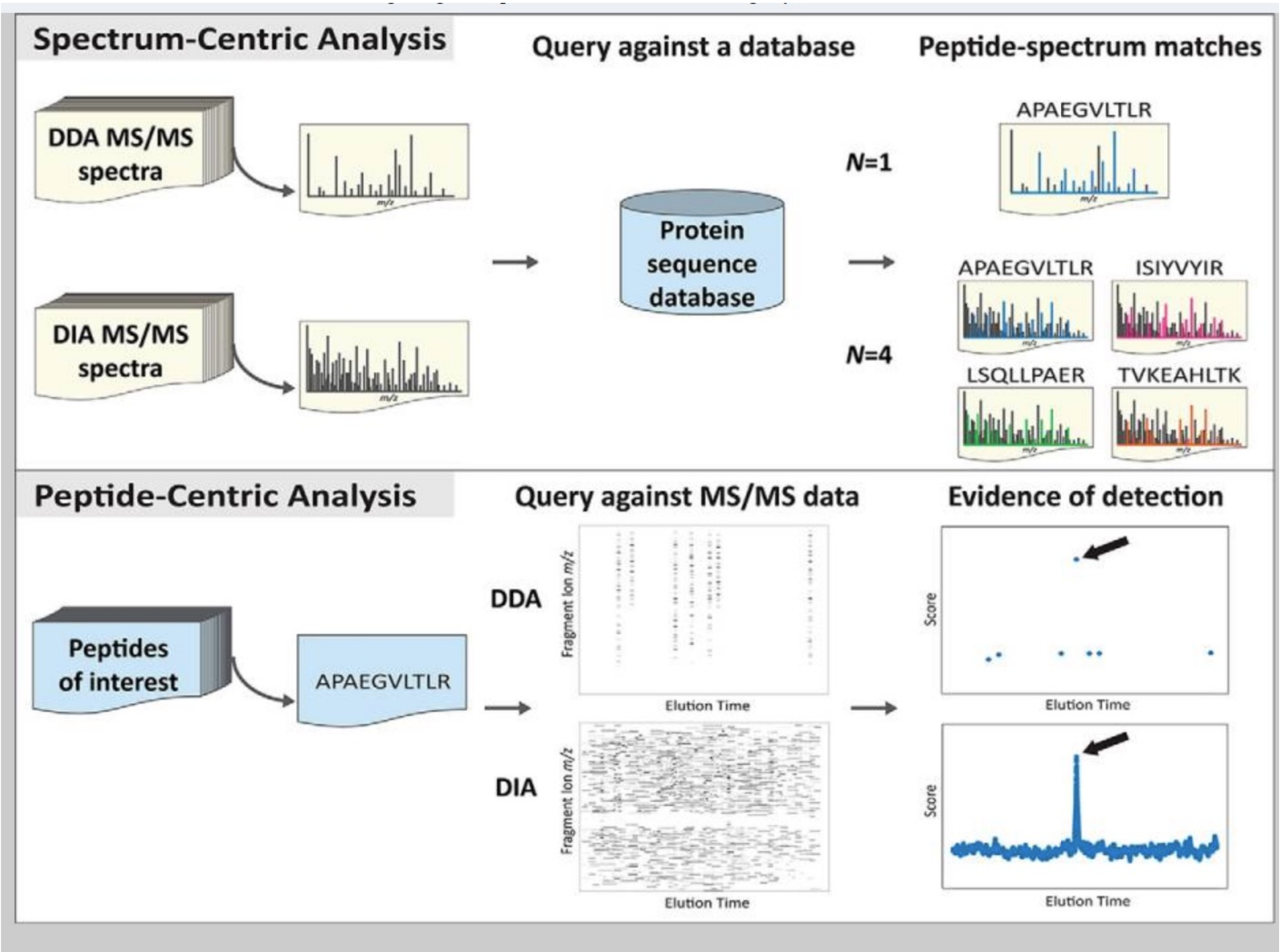
Peptide Centric



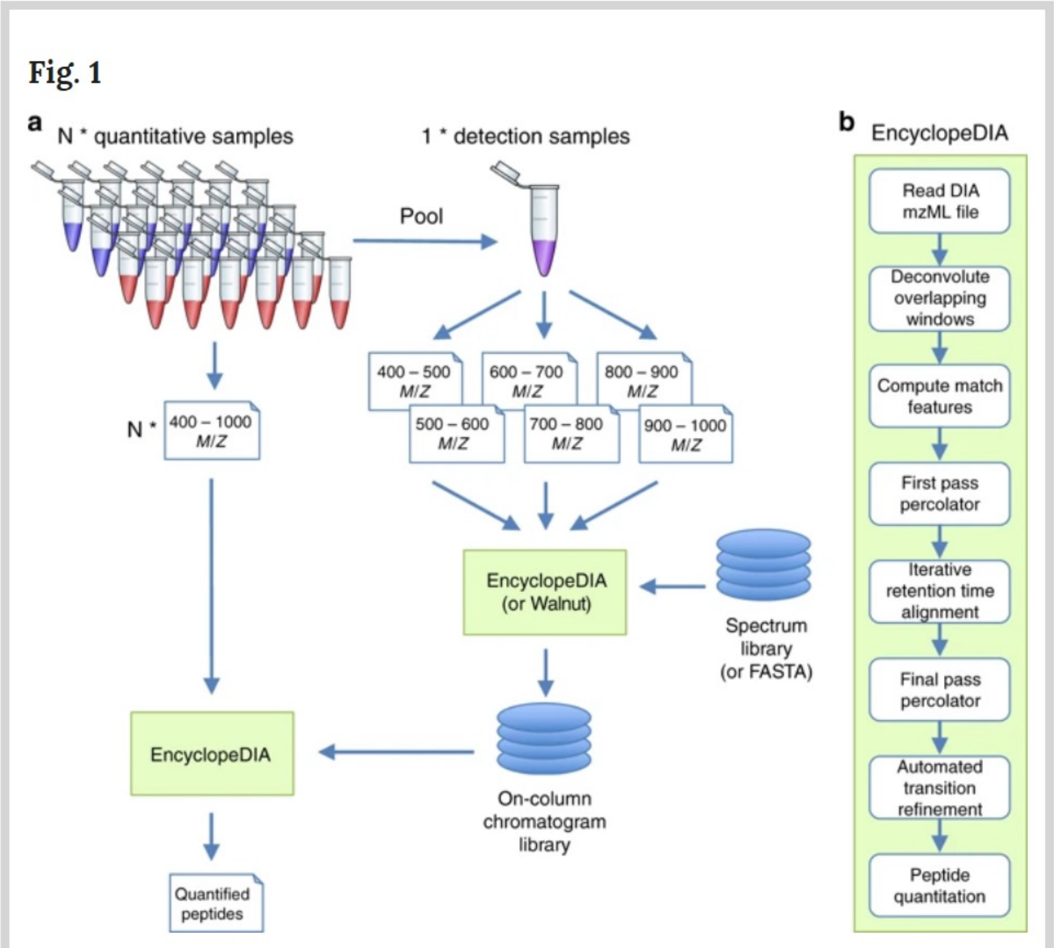




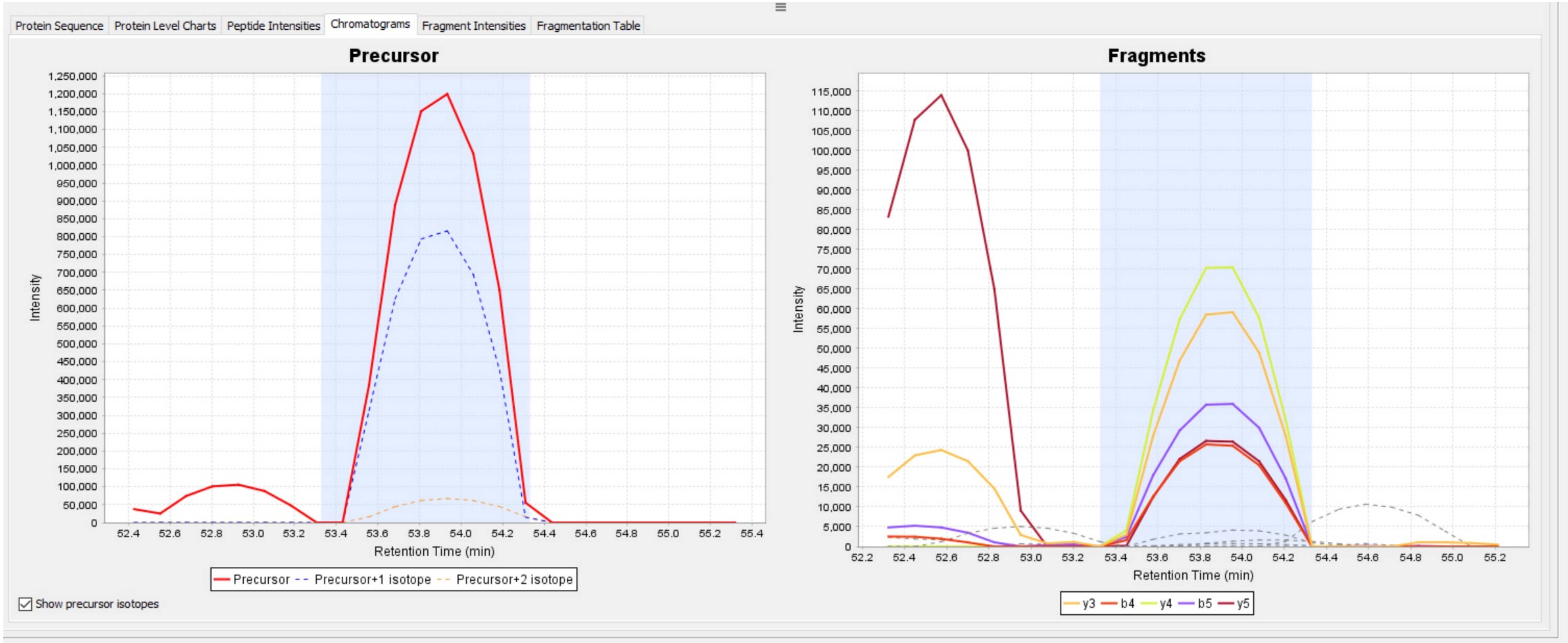
Peptide Centric DIA



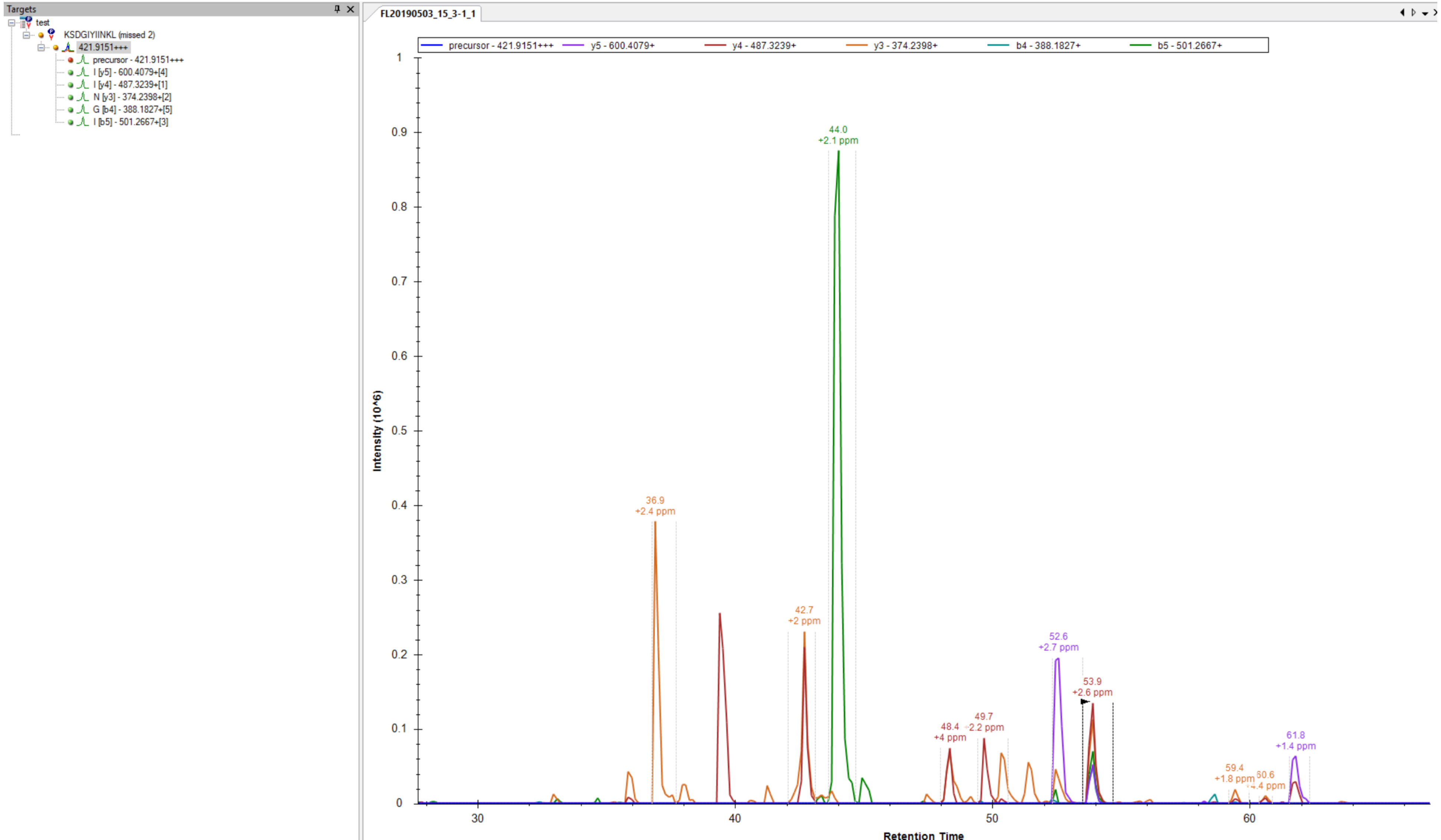
Ting 2015



Searle 2018


Protein Sequence
Protein Level Charts
Peptide Intensities
Chromatograms
Fragment Intensities
Fragmentation Table
Precursor
Intensity
Retention Time (min)
Precursor
Precursor+1 isotope
Precursor+2 isotope
Show precursor isotopes
Fragments
Intensity
Retention Time (min)
y3
b4
y4
b5
y5

DIA Random peptide KSDGIYIINLK



Chromatogram Libraries made with Deep learning!

nature > nature methods > articles > article

nature methods

Article | Published: 27 May 2019

Prosit: proteome-wide prediction of peptide tandem mass spectra by deep learning

Siegfried Gessulat, Tobias Schmidt, Daniel Paul Zolg, Patroklos Samaras, Karsten Schnatbaum, Johannes Zerweck, Tobias Knaute, Julia Rechenberger, Bernard Delanghe, Andreas Huhmer, Ulf Reimer, Hans-Christian Ehrlich, Stephan Aiche, Bernhard Kuster & Mathias Wilhelm

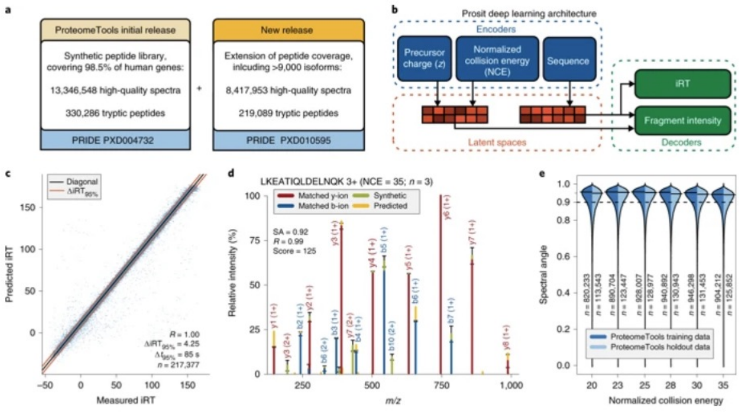
Using Deep Learning to Predict DDA libraries

One issue with peptide DIA is that it seems to works a lot better with a DDA Library

- You need to generate one yourself
- Or use a public one (Pan Human)

- But now, with **Prosit**, you can use deep learning to predict one without generating one using your mass spec!
  - Introduces at ASMS this year

Fig. 1: Accurate retention time and fragment ion intensity prediction by deep learning.



Article | Published: 27 May 2019

Prosit: proteome-wide prediction of peptide tandem mass spectra by deep learning

Siegfried Gessulat, Tobias Schmidt, Daniel Paul Zolg, Patroklos Samaras, Karsten Schnatbaum, Johannes Zerweck, Tobias Knaute, Julia Rechenberger, Bernard Delanghe, Andreas Huhmer, Ulf Reimer, Hans-Christian Ehrlich, Stephan Aiche, Bernhard Kuster & Mathias Wilhelm

Nature Methods **16**, 509–518 (2019) | Download Citation

New Results

Comment on this paper

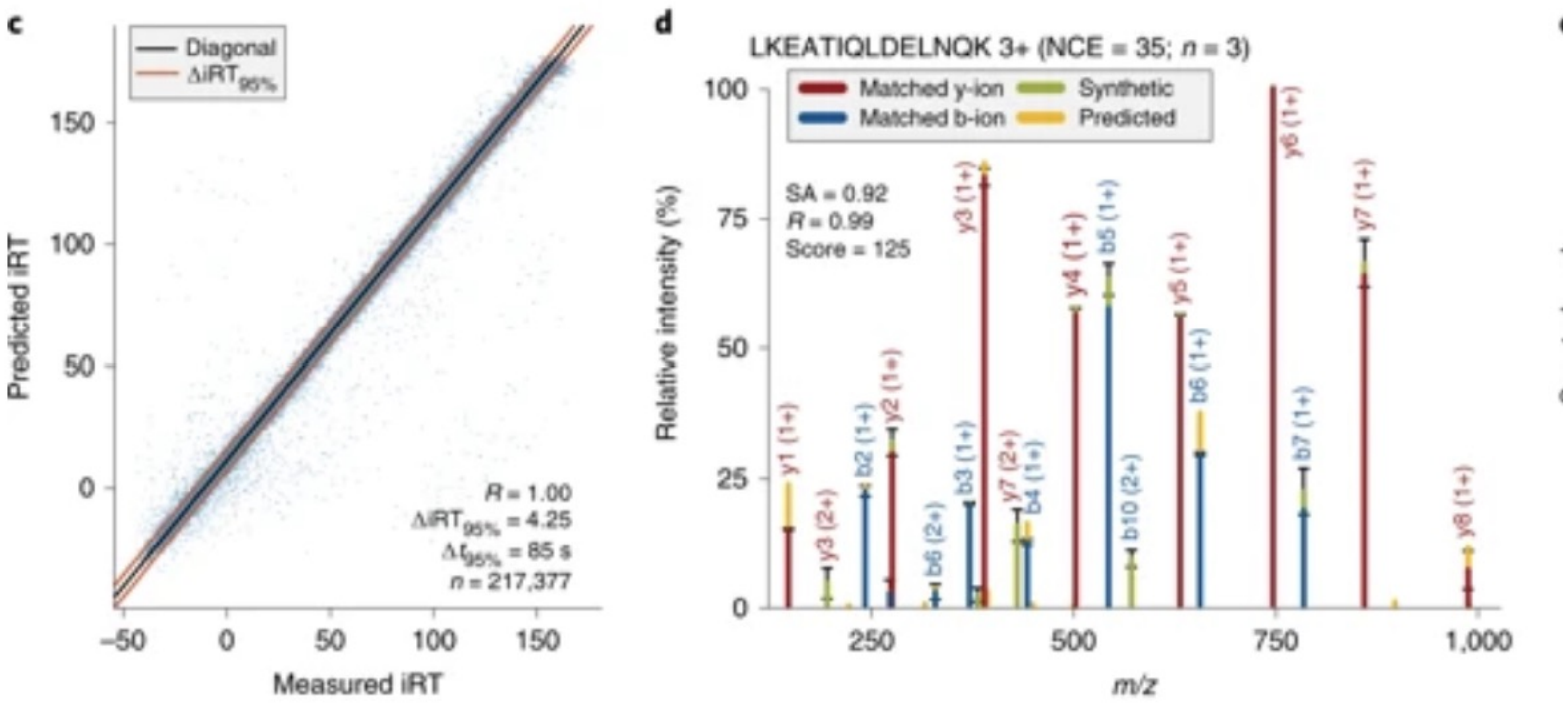
Generating high-quality libraries for DIA-MS with empirically-corrected peptide predictions

Brian C. Searle, Kristian E. Swearingen, Christopher A. Barnes, Tobias Schmidt, Siegfried Gessulat, Bernhard Kuster, Mathias Wilhelm

doi: https://doi.org/10.1101/682245

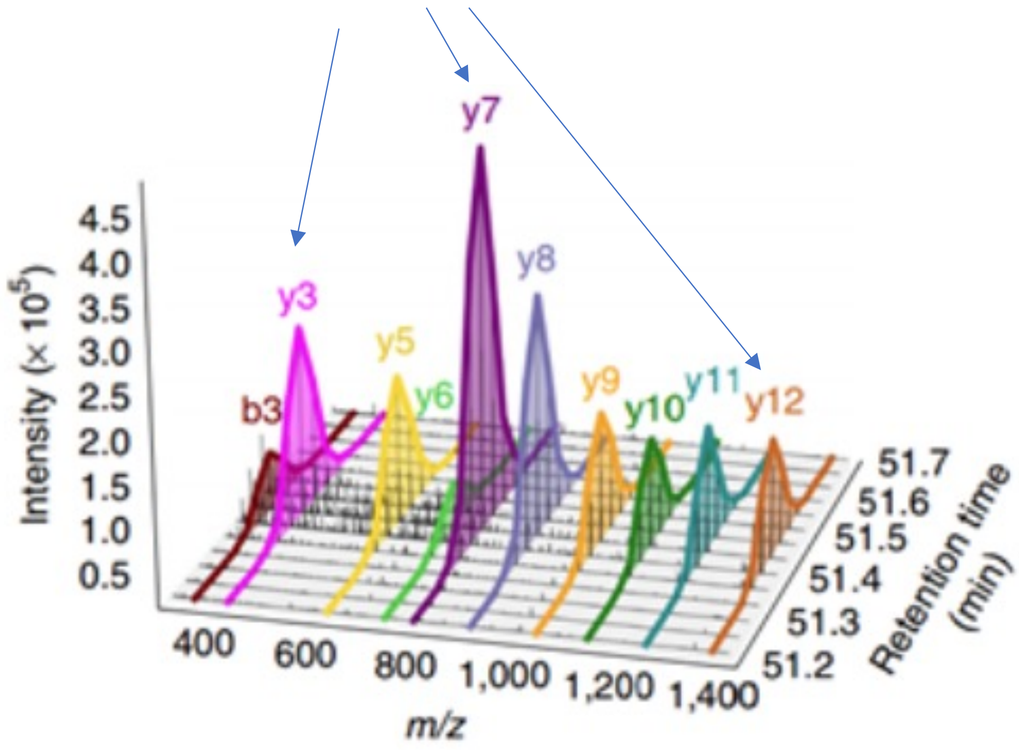
Prosit used millions of synthetic peptides to build deep learning models

- Predicts what y and b ion you should see
- Predicts the intensity of fragment ions
- Predicts retention time

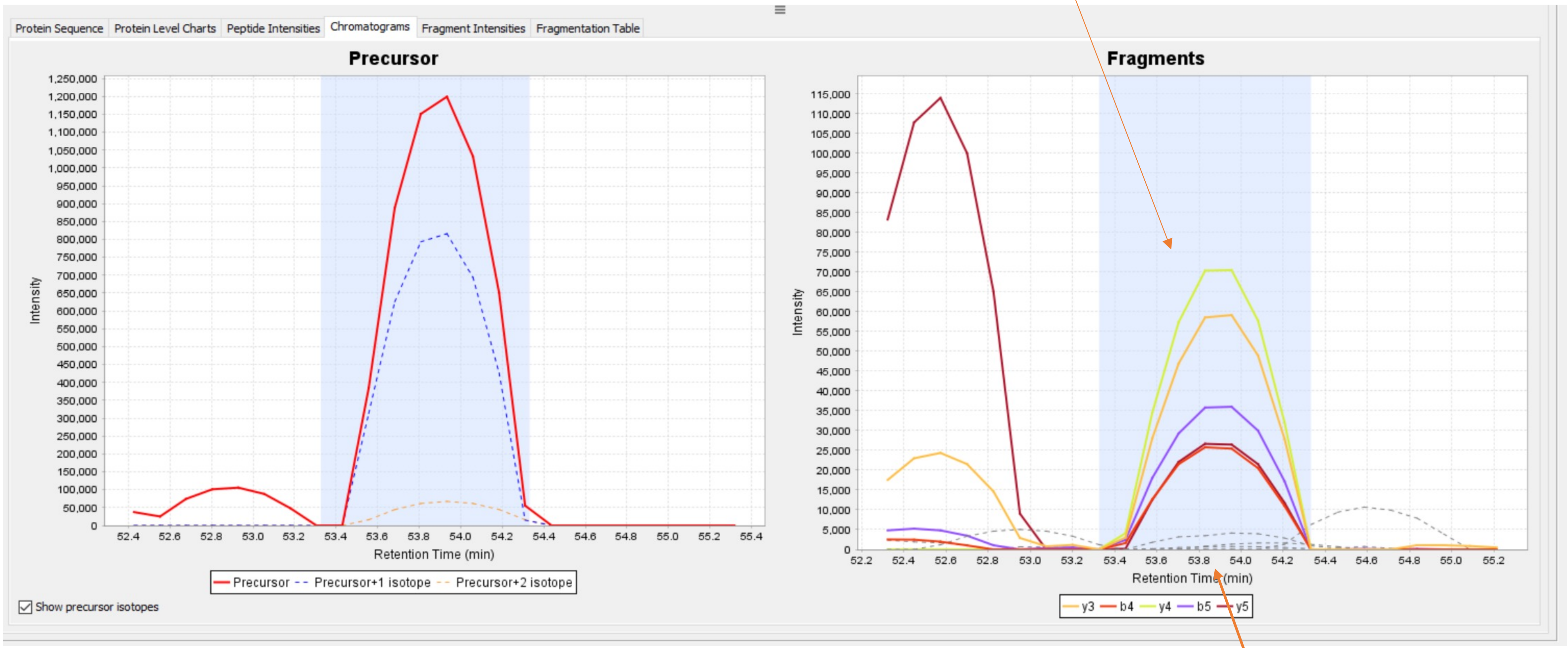


Pros: It works!
Cons: does not tell us anything about the underlying chemistry

These intensities can now be predicted accurately

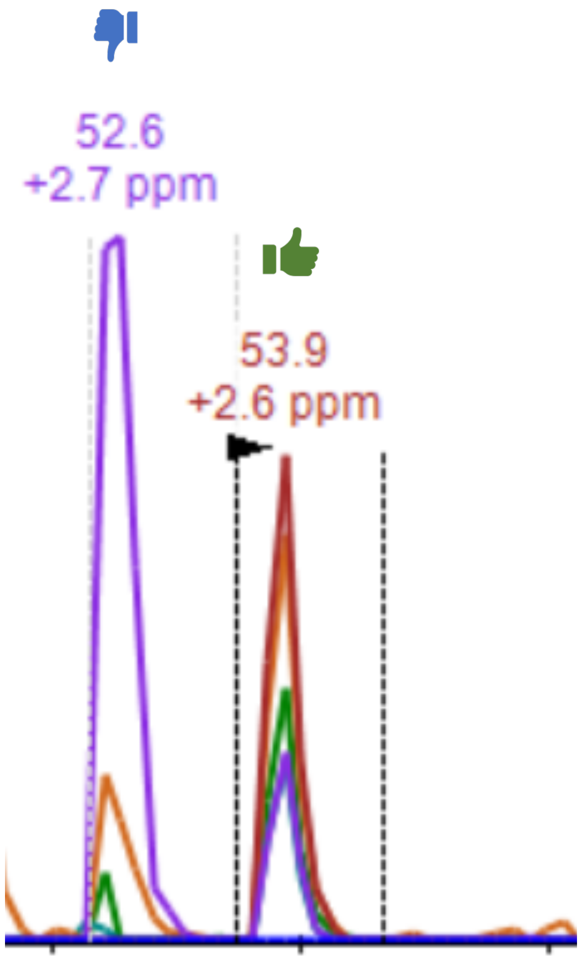


We Can check if these Ratios are right!



We can also use deep learning to check if the RT is right!

DIA Random peptide KSDGIYIINLK



Why is the one on the right matches the predicted ratios the one on the left does not